

PVsyst - Simulation report

Grid-Connected System

Project: batuoncel

Variant: New simulation variant

Tables on a building

System power: 14.28 kWp

Güzelyalı - Turkey



Project: batuoncel

Variant: New simulation variant

PVsyst V8.1.2

VCO, Simulation date:
10/05/26 22:30
with V8.1.2

Project summary

Geographical Site

Güzelyalı
Turkey

Situation

Latitude 38.3979 °(N)
Longitude 27.0846 °(E)
Altitude 5 m
Time zone UTC+3

Project settings

Albedo 0.20

Weather data

Güzelyalı
Meteonorm 9.0 dll, Sat=100% - Synthetic

System summary

Grid-Connected System

Orientation #1

Fixed plane

Tilt/Azimuth 7 / 0 °

System information

PV Array

Nb. of modules 24 units
Pnom total 14.28 kWp

Tables on a building

Orientation #2

Fixed plane

Tilt/Azimuth 7 / 180 °

Near Shadings

Linear shadings : Fast (table)

Inverters

Nb. of units 1 unit
Total power 10 kWac
Pnom ratio 1.43

User's needs

Unlimited load (grid)

Results summary

Simulation step Hourly

Produced Energy 21188 kWh/year Specific production 1484 kWh/kWp/year Perf. Ratio PR 83.44 %

Table of contents

Project and results summary	2
General parameters, PV Array Characteristics, System losses	3
Horizon definition	5
Near shading definition - Iso-shadings diagram	6
Main results	8
Loss diagram	9
Predef. graphs	10
Cost of the system	11
Financial analysis	12
CO ₂ Emission Balance	15



PVsyst V8.1.2

VC0, Simulation date:
10/05/26 22:30
with V8.1.2

General parameters

Grid-Connected System

Tables on a building

Simulation step

Hourly

Orientation #1

Fixed plane

Tilt/Azimuth 7 / 0 °

Sheds configuration

Nb. of sheds 12 units

Set of tables

Shading limit angle

Limit profile angle 82.4 °

Sizes

Sheds spacing 1.14 m

Sensitive width 1.13 m

GCR Shading 99.1 %

Orientation #2

Fixed plane

Tilt/Azimuth 7 / 180 °

Sheds configuration

Nb. of sheds 12 units

Set of tables

Shading limit angle

Limit profile angle 82 °

Sizes

Sheds spacing 1.14 m

Sensitive width 1.13 m

GCR Shading 99.0 %

Models used

Transposition Perez

Diffuse Perez, Meteonorm

Circumsolar separate

Horizon

Average Height 2.9 °

Near Shadings

Linear shadings : Fast (table)

User's needs

Unlimited load (grid)

PV Array Characteristics

PV module

Manufacturer Generic

Model ARCLK-144PV10T-GG-595.v0

(Custom parameters definition)

ARCLK-144PV10T-GG-595.v0.PAN

Unit Nom. Power 595 Wp

Inverter

Manufacturer

Model SUN2000-10KTL-M2-400V

(Original PVsyst database)

Unit Nom. Power 10.00 kWac

Array #1 - PV Array

Orientation 1

Tilt/Azimuth 7/0 °

Number of PV modules 12 units

Nominal (STC) 7.14 kWp

Modules 1 strings x 12 In series

Number of inverters 1 * MPPT 50% 0.5 unit

Total power 5.0 kWac

At operating cond. (50°C)

Pmpp 6.57 kWp

U mpp 477 V

I mpp 14 A

Operating voltage 160-950 V

Max. power (=>60°C) 11.00 kWac

Pnom ratio (DC:AC) 1.43

Array #2 - Sub-array #2

Orientation 2

Tilt/Azimuth 7/180 °

Number of PV modules 12 units

Nominal (STC) 7.14 kWp

Modules 1 strings x 12 In series

Number of inverters 1 * MPPT 50% 0.5 unit

Total power 5.0 kWac

At operating cond. (50°C)

Pmpp 6.57 kWp

U mpp 477 V

I mpp 14 A

Operating voltage 160-950 V

Max. power (=>60°C) 11.00 kWac

Pnom ratio (DC:AC) 1.43



PVsyst V8.1.2

VC0, Simulation date:
10/05/26 22:30
with V8.1.2

PV Array Characteristics

Total PV power		Total inverter power	
Nominal (STC)	14 kWp	Total power	10 kWac
Total	24 modules	Number of inverters	1 unit
Module area	62.0 m²	Pnom ratio	1.43
Cell area	57.2 m²	No power sharing	

Array losses

Thermal Loss factor

Module temperature according to irradiance

Uc (const) 20.0 W/m²K

Uv (wind) 0.0 W/m²K/m/s

DC wiring losses

Global array res. 570 mΩ

Global wiring resistance 285 mΩ

Loss Fraction 1.50 % at STC

LID - Light Induced Degradation

Loss Fraction 2.00 %

Module Quality Loss

Loss Fraction -0.75 %

Module mismatch losses

Loss Fraction 2.00 % at MPP

IAM loss factor

Incidence effect (IAM): Fresnel, AR coating, n(glass)=1.526, n(AR)=1.290

0°	30°	50°	60°	70°	75°	80°	85°	90°
1.000	0.999	0.987	0.963	0.892	0.814	0.679	0.438	0.000



PVsyst V8.1.2

VC0, Simulation date:
10/05/26 22:30
with V8.1.2

Horizon definition

Horizon from PVGIS website API, Lat=38°23'52", Long=27°5'4", Alt=5m

Average Height	2.9 °	Albedo Factor	0.92
Diffuse Factor	1.00	Albedo Fraction	100 %

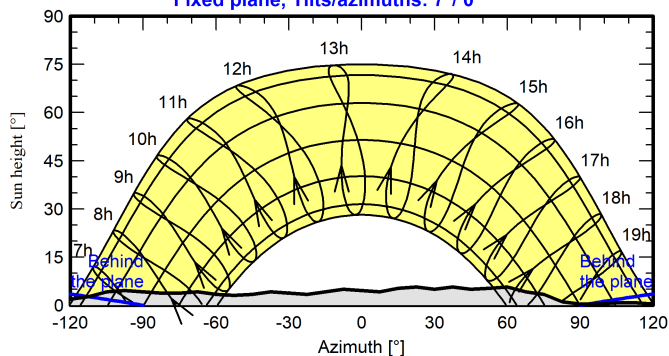
Horizon profile

Azimuth [°]	-180	-173	-165	-158	-150	-143	-135	-128	-120	-113	-105	-98
Height [°]	1.5	2.3	2.3	2.7	3.4	2.7	1.9	1.5	1.9	2.7	4.2	4.6
Azimuth [°]	-90	-83	-75	-68	-60	-53	-45	-38	-30	-23	-15	-8
Height [°]	4.2	3.8	3.8	4.2	3.4	3.1	3.4	4.2	3.8	3.4	4.2	5.0
Azimuth [°]	8	15	23	30	38	45	53	60	68	75	83	90
Height [°]	4.2	5.3	5.7	5.0	5.7	5.0	5.3	5.7	4.2	3.4	1.1	0.4
Azimuth [°]	98	105	113	120	135	143	165	173	180			
Height [°]	0.4	0.8	0.8	0.4	0.4	0.8	0.8	1.1	1.5			

Sun Paths (Height / Azimuth diagram)

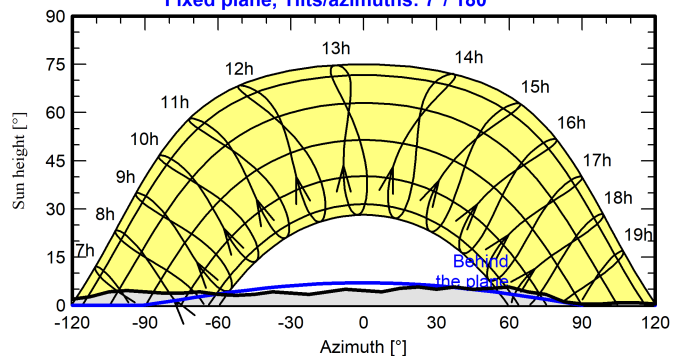
Orientation #1

Fixed plane, Tilts/azimuths: 7°/ 0°



Orientation #2

Fixed plane, Tilts/azimuths: 7°/ 180°



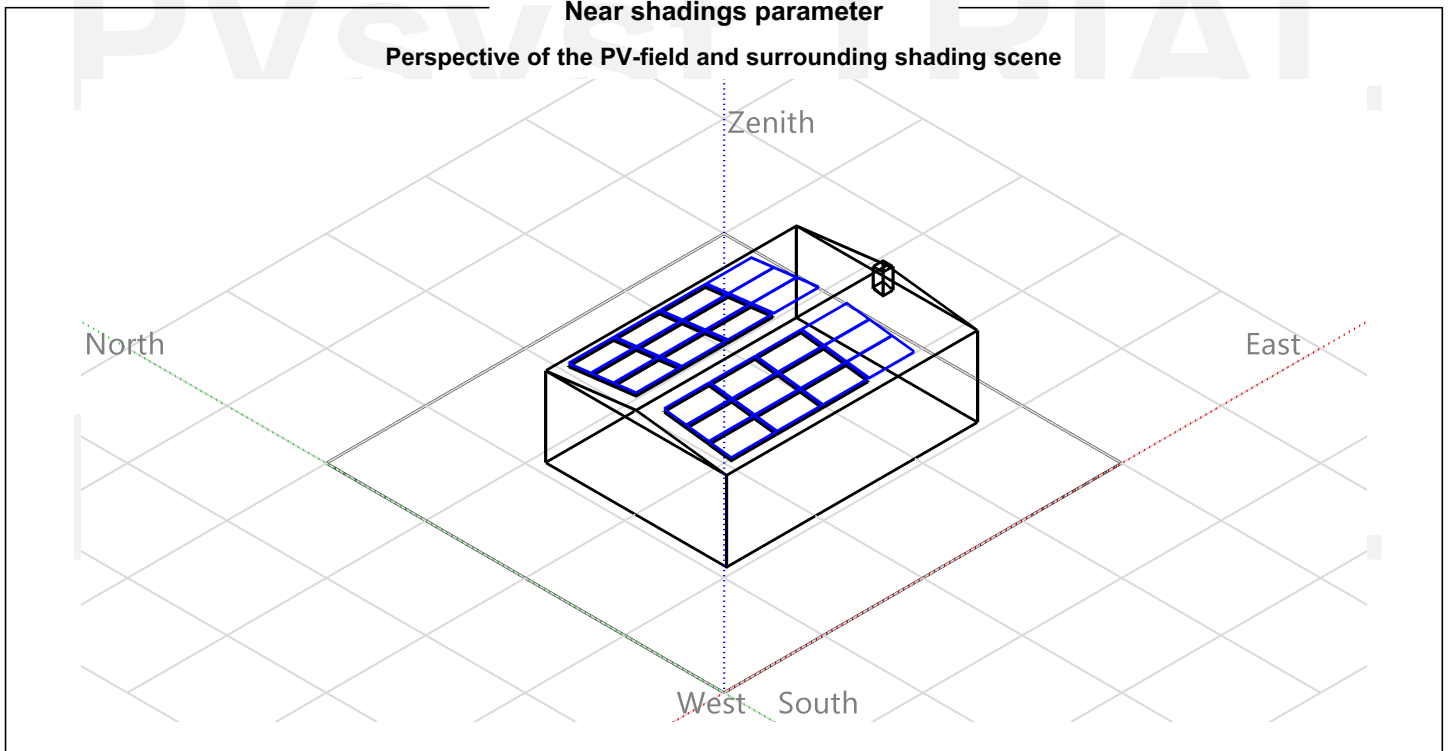


PVsyst V8.1.2

VC0, Simulation date:
10/05/26 22:30
with V8.1.2

Near shadings parameter

Perspective of the PV-field and surrounding shading scene





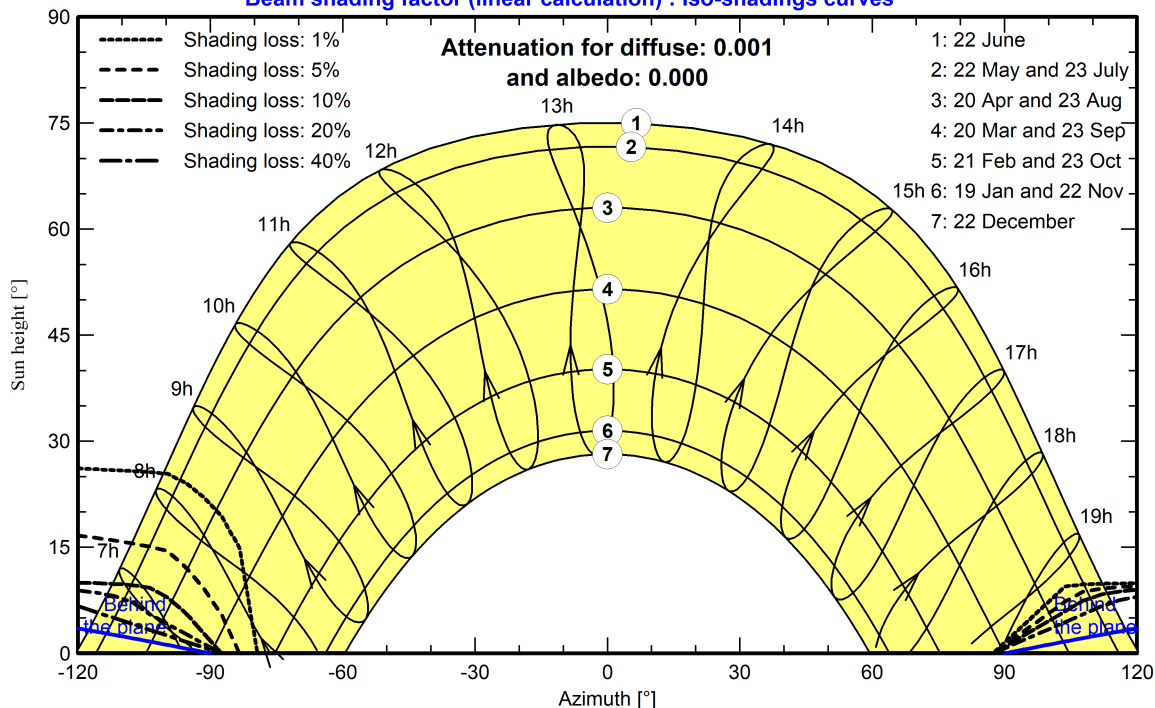
PVsyst V8.1.2

VC0, Simulation date:
10/05/26 22:30
with V8.1.2

Iso-shadings diagram

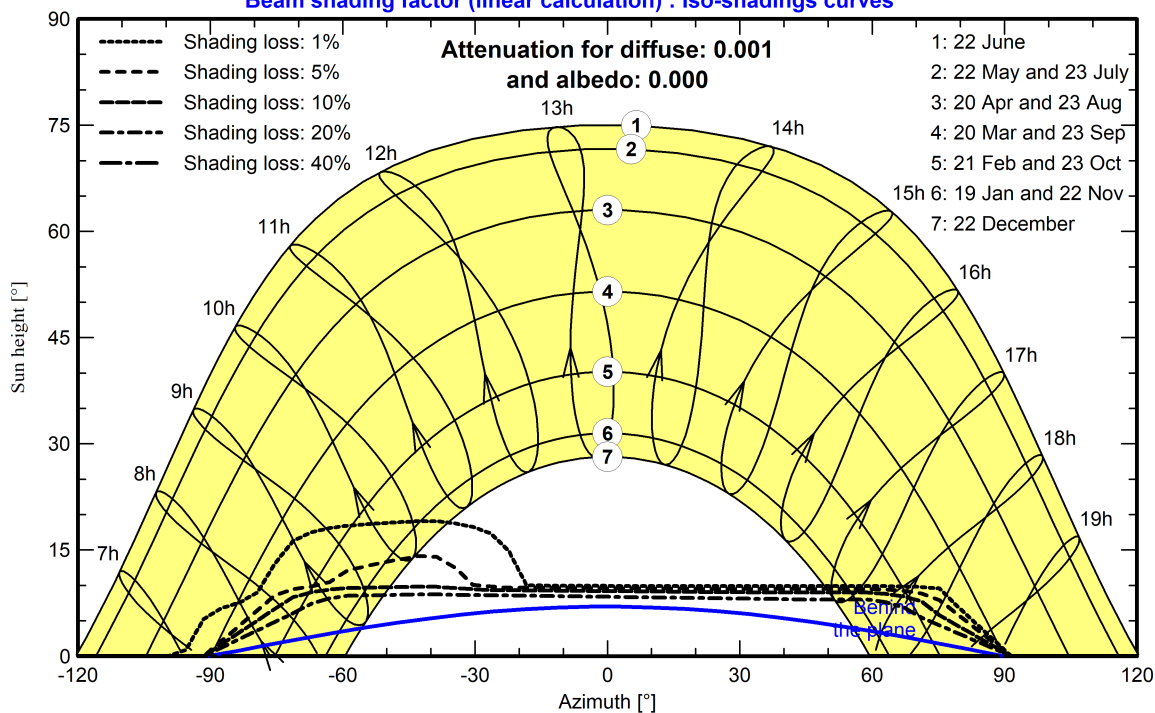
Orientation #1 - Fixed plane, Tilts/azimuths: 7°/ 0°

Beam shading factor (linear calculation) : Iso-shadings curves



Orientation #2 - Fixed plane, Tilts/azimuths: 7°/ 180°

Beam shading factor (linear calculation) : Iso-shadings curves





Project: batuoncel
Variant: New simulation variant

PVsyst V8.1.2

VC0, Simulation date:
10/05/26 22:30
with V8.1.2

Main results

Simulation step

Hourly

System Production

Produced Energy 21188 kWh/year Specific production 1484 kWh/kWp/year
Perf. Ratio PR 83.44 %

Economic evaluation

Investment

Global 10.080.00 USD
Specific 0.71 USD/Wp

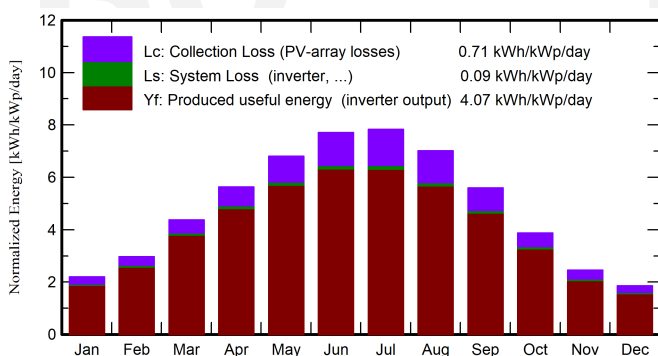
Yearly cost

Annuities
Run. costs
Payback period

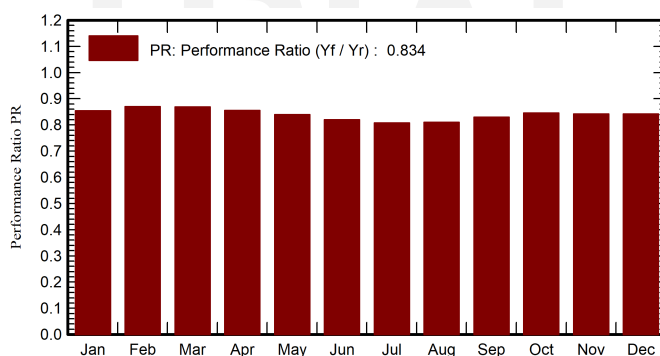
LCOE

0.00 USD/yr Energy cost 0.02 USD/kWh
0.00 USD/yr
3.4 years

Normalized productions (per installed kWp)



Performance Ratio PR



Balances and main results

	GlobHor	DiffHor	T_Amb	GlobInc	GlobEff	EArray	E_Grid	PR
	kWh/m ²	kWh/m ²	°C	kWh/m ²	kWh/m ²	kWh	kWh	ratio
January	68.4	33.17	8.42	68.2	63.1	857	832	0.855
February	83.2	38.44	10.05	83.1	78.7	1059	1032	0.870
March	136.1	57.71	12.92	135.7	130.8	1723	1684	0.869
April	169.4	66.19	16.66	169.1	164.1	2110	2065	0.855
May	211.4	78.02	21.53	211.0	205.6	2582	2529	0.839
June	231.7	70.40	26.05	231.3	225.7	2767	2709	0.820
July	243.4	59.58	28.81	242.8	236.9	2859	2799	0.807
August	217.9	54.28	28.77	217.4	211.6	2571	2517	0.811
September	168.1	48.75	24.21	167.8	162.1	2032	1988	0.830
October	120.5	41.72	18.98	120.3	114.8	1487	1451	0.845
November	74.0	32.73	14.34	73.8	68.8	913	888	0.842
December	57.9	27.68	10.12	57.7	53.1	716	693	0.842
Year	1782.1	608.68	18.45	1778.2	1715.4	21675	21188	0.834

Legends

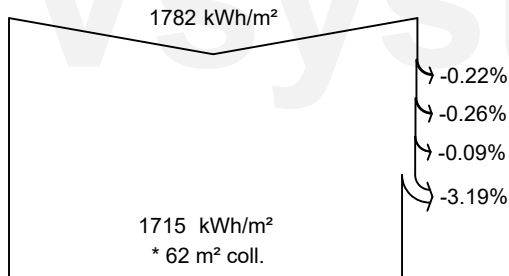
GlobHor Global horizontal irradiation
DiffHor Horizontal diffuse irradiation
T_Amb Ambient Temperature
GlobInc Global incident in coll. plane
GlobEff Effective Global, corr. for IAM and shadings
EArray Effective energy at the output of the array
E_Grid Energy injected into grid
PR Performance Ratio



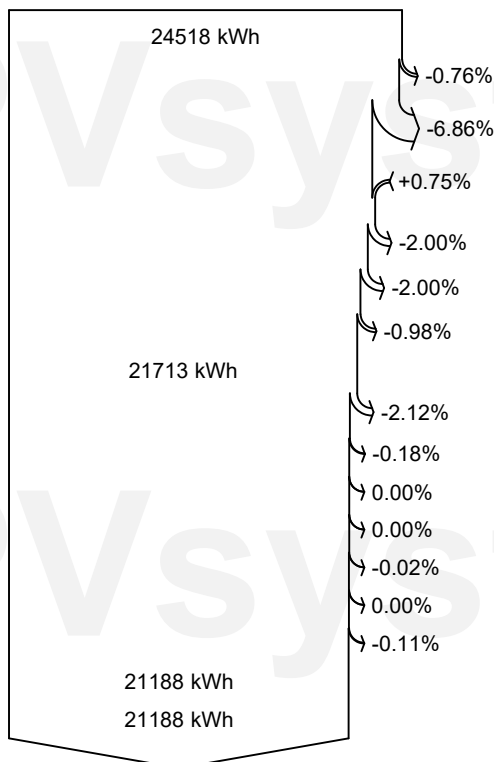
PVsyst V8.1.2

VC0, Simulation date:
10/05/26 22:30
with V8.1.2

Loss diagram



efficiency at STC = 23.05%



Global horizontal irradiation

Global incident in coll. plane

Far Shadings / Horizon

Near Shadings: irradiance loss

IAM factor on global

Effective irradiation on collectors

PV conversion

Array nominal energy (at STC effic.)

PV loss due to irradiance level

PV loss due to temperature

Module quality loss

LID - Light induced degradation

Module array mismatch loss

Ohmic wiring loss

Array virtual energy at MPP

Inverter Loss during operation (efficiency)

Inverter Loss over nominal inv. power

Inverter Loss due to max. input current

Inverter Loss over nominal inv. voltage

Inverter Loss due to power threshold

Inverter Loss due to voltage threshold

Night consumption

Available Energy at Inverter Output

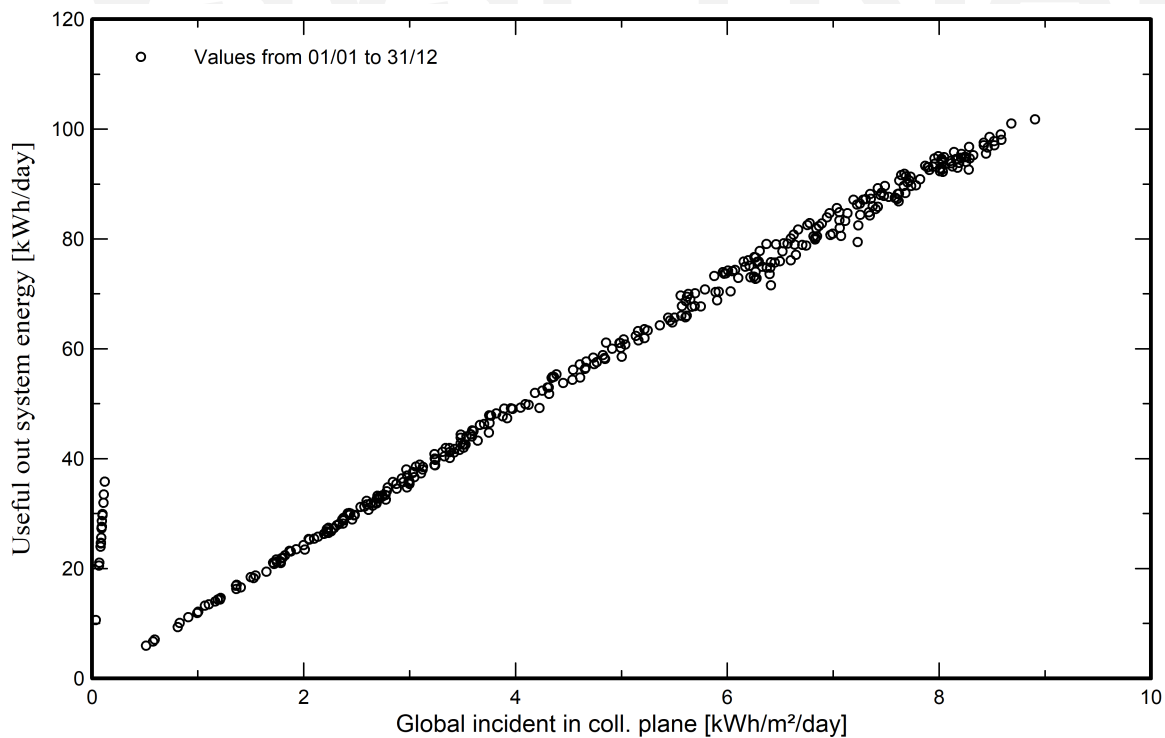
Energy injected into grid



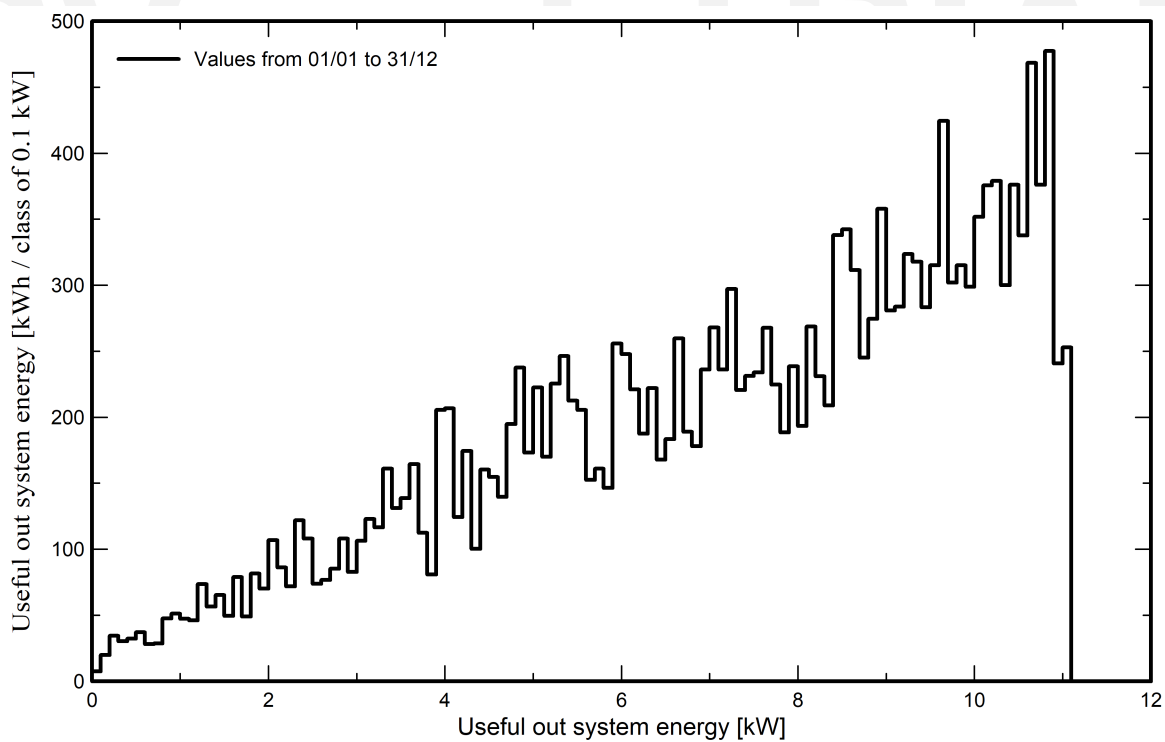
PVsyst V8.1.2

VC0, Simulation date:
10/05/26 22:30
with V8.1.2

Predef. graphs
Daily Input/Output diagram



System Output Power Distribution





PVsyst V8.1.2

VC0, Simulation date:
10/05/26 22:30
with V8.1.2

Cost of the system

Installation costs

Item	Quantity units	Cost USD	Total USD
PV modules ARCLK-144PV10T-GG-595.v0	24	120.00	2.880.00
Inverters SUN2000-10KTL-M2-400V	1	1.500.00	1.500.00
Studies and analysis Engineering	1	5.700.00	5.700.00
		Total	10.080.00
		Depreciable asset	4.380.00

Operating costs

Item	Total USD/year
Total (OPEX)	0.00
Including inflation (4.00%)	0.00

System summary

Total installation cost	10.080.00 USD
Operating costs (incl. inflation 4.00%/year)	0.00 USD/year
Produced Energy	21.2 MWh/year
Cost of produced energy (LCOE)	0.0238 USD/kWh



PVsyst V8.1.2

VC0, Simulation date:
10/05/26 22:30
with V8.1.2

Financial analysis

Simulation period

Project lifetime 20 years Start year 2027

Income variation over time

Inflation 4.00 %/year
Module Degradation 0.00 %/year
Discount rate 0.00 %/year

Income dependent expenses

Income tax rate 0.00 %/year
Other income tax 0.00 %/year
Dividends 0.00 %/year

Depreciable assets

Asset	Depreciation method	Depreciation period (years)	Salvage value (USD)	Depreciable (USD)
PV modules ARCLK-144PV10T-GG-595.v0	Straight-line	20	0.00	2.880.00
Inverters SUN2000-10KTL-M2-400V	Straight-line	20	0.00	1.500.00
		Total	0.00	4.380.00

Financing

Own funds 10.080.00 USD

Electricity sale

Feed-in tariff 0.14000 USD/kWh
Duration of tariff warranty 20 years
Annual connection tax 0.00 USD/year
Annual tariff variation 0.0 %/year
Feed-in tariff decrease after warranty 0.00 %

Return on investment

Payback period 3.4 years
Net present value (NPV) 49.311.81 USD
Internal rate of return (IRR) 29.29 %
Return on investment (ROI) 489.2 %



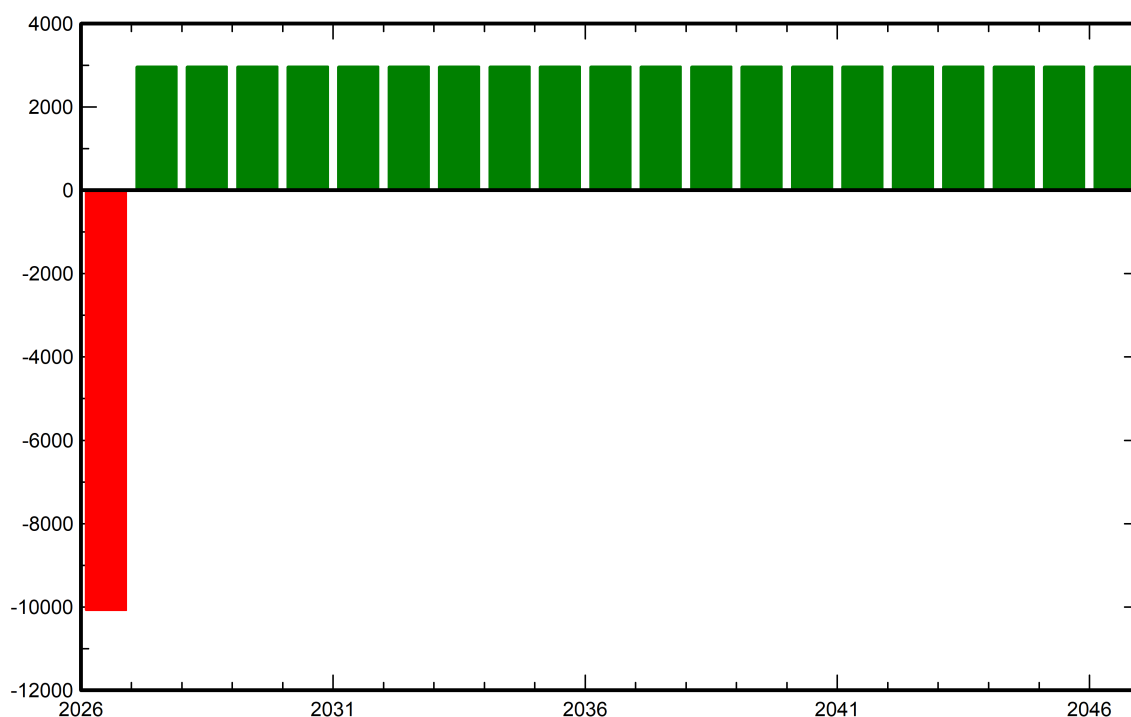
PVsyst V8.1.2

VC0, Simulation date:
10/05/26 22:30
with V8.1.2

Financial analysis
Detailed economic results (USD)

Year	Electricity sale	Own funds	Run. costs	Deprec. allow.	Taxable income	Taxes	After-tax profit	Cumul. profit	% amorti.
0	0	10.080	0	0	0	0	0	-10.080	0.0%
1	2.970	0	0	219	2.751	0	2.970	-7.110	29.5%
2	2.970	0	0	219	2.751	0	2.970	-4.141	58.9%
3	2.970	0	0	219	2.751	0	2.970	-1.171	88.4%
4	2.970	0	0	219	2.751	0	2.970	1.798	117.8%
5	2.970	0	0	219	2.751	0	2.970	4.768	147.3%
6	2.970	0	0	219	2.751	0	2.970	7.738	176.8%
7	2.970	0	0	219	2.751	0	2.970	10.707	206.2%
8	2.970	0	0	219	2.751	0	2.970	13.677	235.7%
9	2.970	0	0	219	2.751	0	2.970	16.646	265.1%
10	2.970	0	0	219	2.751	0	2.970	19.616	294.6%
11	2.970	0	0	219	2.751	0	2.970	22.585	324.1%
12	2.970	0	0	219	2.751	0	2.970	25.555	353.5%
13	2.970	0	0	219	2.751	0	2.970	28.525	383.0%
14	2.970	0	0	219	2.751	0	2.970	31.494	412.4%
15	2.970	0	0	219	2.751	0	2.970	34.464	441.9%
16	2.970	0	0	219	2.751	0	2.970	37.433	471.4%
17	2.970	0	0	219	2.751	0	2.970	40.403	500.8%
18	2.970	0	0	219	2.751	0	2.970	43.373	530.3%
19	2.970	0	0	219	2.751	0	2.970	46.342	559.7%
20	2.970	0	0	219	2.751	0	2.970	49.312	589.2%
Total	59.392	10.080	0	4.380	55.012	0	59.392	49.312	589.2%

Yearly net profit (USD)

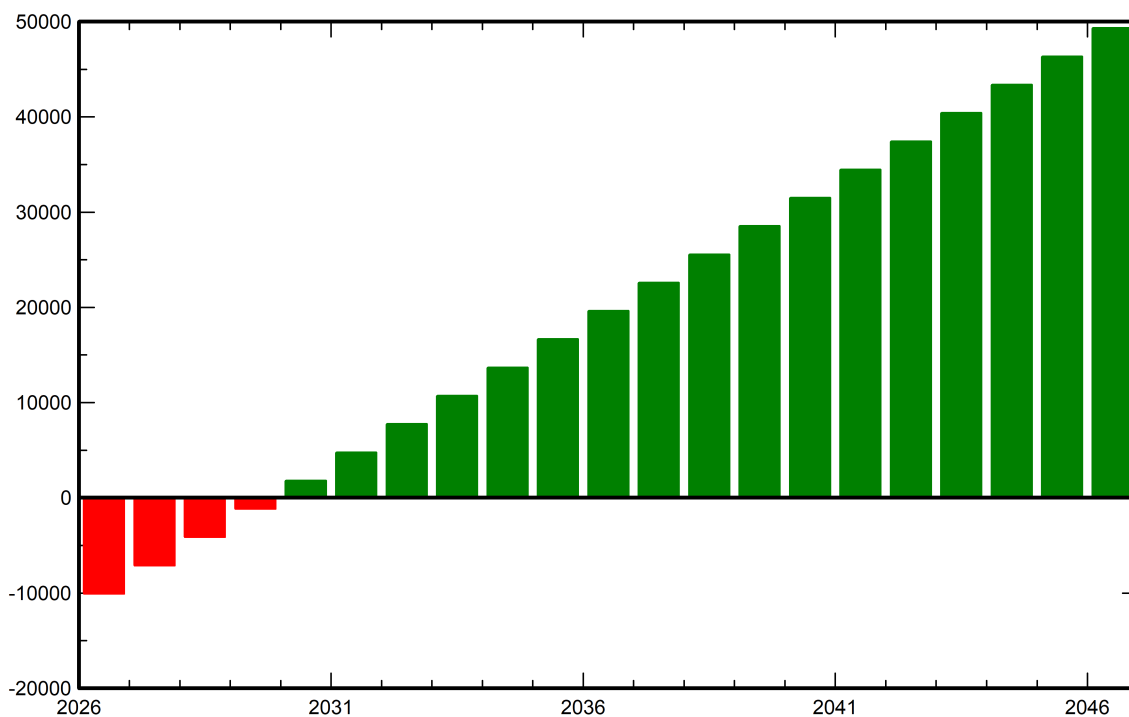




PVsyst V8.1.2

VC0, Simulation date:
10/05/26 22:30
with V8.1.2

Financial analysis
Cumulative cashflow (USD)





PVsyst V8.1.2

VC0, Simulation date:
10/05/26 22:30
with V8.1.2

CO₂ Emission Balance

Total: 244.1 tCO₂

Generated emissions

Total: 25.56 tCO₂

Source: Detailed calculation from table below

Replaced Emissions

Total: 310.8 tCO₂

System production: 21.19 MWh/yr

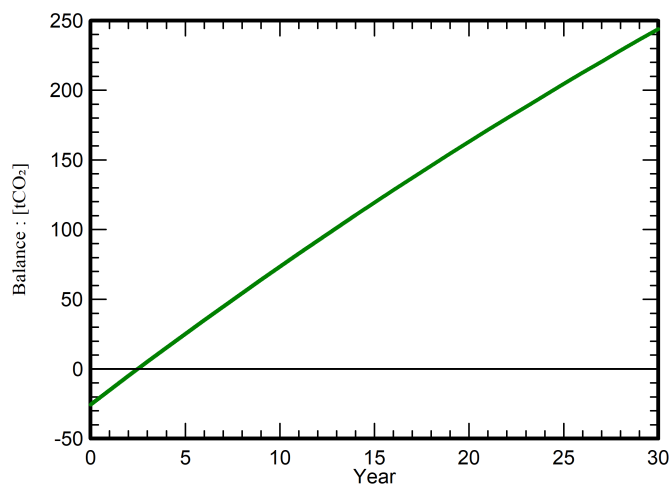
Grid Lifecycle Emissions: 489 gCO₂/kWh

Source: IEA List

Country: Turkey

Lifetime: 30 years

Annual degradation: 1.0 %

Saved CO₂ Emission vs. Time**System Lifecycle Emissions Details**

Item	LCE	Quantity	Subtotal
[kgCO ₂]			
Modules	1713 kgCO ₂ /kWp	14.3 kWp	24458
Supports	3.26 kgCO ₂ /kg	240 kg	783
Inverters	323 kgCO ₂ /units	1.00 units	323